

How You Speak and Write: Harvard Professor Explains The Rules of Writing - Steven Pinker

Various speakers

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Chapter 1

Harvard Professor Explains The Rules of Writing

This lecture begins with a practical promise. We are not asked to enter a theory of language for its own sake; we are asked how to write well, and how to keep writing well in an age in which large language models can already produce smooth, competent prose. Pinker answers that question not with a bag of tips but with a sequence of causal claims, contrasts, and worked examples. The lecture contains no literal blackboard mathematics, so the displayed formulas below are schematic reconstructions of mechanisms stated in prose. They are meant to keep the logic sharp while remaining faithful to the transcript.

1.1 Bad Writing and the Curse of Knowledge

The interviewer opens at once with the governing problem: why is there so much bad writing? Pinker first clears away the most flattering bad answer. It is tempting to say that opaque prose is mainly strategic, that writers hide thin ideas behind jargon in order to signal sophistication, protect a clique, or exclude outsiders. Pinker does not deny that this can happen. But he insists that it is not the main explanation. He knows too many brilliant people with plenty to say who nevertheless produce unreadable prose.

That reversal matters, because it sets the lecture on the right track. The central failure is not moral theater. It is miscalibration.

Definition 1.1. The curse of knowledge is the difficulty of reconstructing what it is like not to know what we ourselves know.

Pinker compresses the failure into a question a writer ought to ask but often cannot answer in time: What does the audience know, and what do they not know? If we turn that into a schematic mechanism, we get

$$\text{expert knowledge} \rightarrow \text{failure to model the audience} \rightarrow \text{jargon / acronyms / abstractions} \rightarrow \text{reader confusion.} \quad (1.1)$$

Remark 1.2. This display is editorial shorthand, not lecture-visible notation. Its purpose is simply to make Pinker's causal chain explicit.

The lecture then earns the formula by example. A brilliant molecular biologist gives a TED talk exactly as if he were speaking to fellow molecular biologists. He does not begin with the problem. He does not explain why the work matters. He launches directly into the experiments. The room is lost almost immediately, and the only person who does not see that is the speaker himself. Pinker's point is severe but precise: the man is not stupid, except in one local and disastrous way. He does not know what it is like not to know what he knows.

The same failure appears when abstraction outruns explanation. Pinker cites the sentence "the level of the stimulus was proportional to the intensity of the reaction." What it actually means, in the example he gives, is that children look longer at a bunny than at a truck. The first sentence is not necessarily false, but it withholds the scene. The second gives the reader something to see.

1.1.1 Question & Answer

Question. Why is there so much bad writing if the writers are often intelligent and sincere?

Answer. Because intelligence in a field does not guarantee skill at locating the reader. The writer's knowledge outruns the writer's model of the audience. Pinker's central move is to replace a story about vanity or malice with a story about failed perspective-taking.

Proposition 1.3. *The dominant failure mode in explanatory prose is not lack of ideas but failure to estimate what the reader can reasonably infer.*

That is why Pinker links the curse of knowledge to egocentrism and to a missing theory of mind. The lecture begins, then, with a cognitive mistake that will govern every later question about examples, structure, rhythm, and style.

1.2 Audience Calibration as a Writing Practice

Once the diagnosis has been stated, the lecture immediately asks for a remedy. If bad writing is a failed audience model, then what do we do while drafting? Pinker's first answer is modest. We try to imagine the reader. We cultivate empathy. But he adds, almost at once, that this is not enough. The curse of knowledge is treacherous precisely because one does not feel it from the inside. What seems too obvious to explain often turns out not to be obvious at all.

Schematically, the revision process looks like

$$\text{draft} \rightarrow \text{imagined reader}, \quad (1.2)$$

$$\text{draft} + \text{actual reader response} \rightarrow \text{revision}. \quad (1.3)$$

The first arrow is necessary. The second is what saves us.

Pinker is especially good here because he refuses a crude picture of the reader. He did not show drafts to his mother because she was dull or unsophisticated; he says the opposite. She was intelligent, well read, and serious. What mattered was that she was not a specialist in his field. The relevant test reader is not a random member of the population, and not a peer. It is someone intelligent enough to follow the argument once it has been properly stated, but distant enough not to share the writer's local shorthand.

The lecture moves through a concrete sequence of readers:

- the writer’s own attempted act of empathy,
- a trusted intelligent non-specialist,
- the editor at a commercial press,
- academics from neighboring fields,
- even students or colleagues who stand just outside the writer’s tiny research circle.

That last point is important and easy to lose. Pinker notes that even inside the academy, and sometimes inside the same department, people become unintelligible to one another because five or six people in a lab have been breathing the same jargon for months. As soon as the prose steps outside that microclimate, it collapses.

So the lecture’s first practical rule is not merely “know your audience.” It is stricter:

Try to get into the reader’s head, but do not trust that effort by itself. Put the draft in front of an actual human being who is smart, curious, and outside the immediate circle.

This is the first major repair mechanism. The second now follows naturally.

1.3 Concreteness, Imagery, and Sensory Understanding

The lecture pivots here in a way that matters for the whole chapter. Once Pinker has spoken about audience calibration, he asks what understanding itself consists in. His answer is that language, for all its importance, is somewhat overrated. We live in language as writers, and Pinker studies language professionally, but understanding is not just the reception of one verbal string after another. Language is a means to get the reader to grasp an idea, and that idea is often visual, motoric, bodily, emotional, auditory, or otherwise sensory.

We may compress that claim into

$$\text{language} \rightarrow \text{mental model}, \quad \text{not merely} \quad \text{language} \rightarrow \text{more language.} \quad (1.4)$$

This is Pinker’s second major rule. If the first rule is to locate the reader, the second is to give the reader something to build. Do not say “stimulus” if you mean “bunny.” Do not hide behind level, perspective, framework, paradigm, or concept when what the reader needs is a scene.

A worked example. The lecture supplies a near-perfect miniature derivation from abstraction to observability:

1. Begin with the sentence “the level of the stimulus was proportional to the intensity of the reaction.”
2. Notice that the sentence records a relation while suppressing almost all of the perceptible content.
3. Ask what the reader is supposed to picture.
4. Restore the concrete entities and action: children, bunny, truck, looking longer.

5. The claim becomes visible, and therefore thinkable, as prose.

Written schematically,

“The level of the stimulus was proportional to the intensity of the reaction.” $\nRightarrow$ a stable scene for the reader,
(1.5)

“Kids look longer at a bunny than at a truck.” $\Rightarrow$ an inspectable mental picture.
(1.6)

The lecture does not stop with this one example. It broadens outward to visual metaphor. Earlier prose often strikes us as more vivid, Pinker suggests, because earlier writers had fewer ready-made abstractions available and therefore had to lean on images held in common. He gives the striking phrase “the spirit of the hawk kneaded into our flesh” as the sort of expression that can do the work that a later technical term like “aggression” or “antisocial behavior” now performs. We need not imitate the older style wholesale, but we should understand the mechanism: the writer reaches the reader by presenting something imaginable.

1.4 Why Writing Is Harder Than Speaking

At this point the lecture raises a fresh obstacle. If speech comes so naturally to children, why does writing remain slow, effortful, and artificial? Pinker answers by changing level. Up to now he has been offering advice. Here he explains the environment in which the advice becomes necessary.

1.4.1 Question & Answer

Question. Why does writing feel unnatural compared with speech?

Answer. Because conversation begins inside a shared situation, while writing must construct its own situation out of marks on a page.

The lecture’s contrast can be written, again schematically, as

$$\text{conversation} = \text{common ground} + \text{deixis} + \text{feedback}, \quad (1.7)$$

$$\text{writing} = \text{text on the page}. \quad (1.8)$$

This small pair of equations condenses several distinct points in the transcript:

1. Conversation does not begin from nowhere. The participants know why they are there and what occasion brought them together.
2. Because the context is shared, words like “this,” “that,” “she,” “the thing,” and “what I was talking about” are often perfectly clear.
3. Conversation supplies immediate correction: the furrowed brow, the puzzled glance, the request for clarification, the drift of attention.
4. Even live public speaking retains some of this support, because the speaker can still watch the audience.

5. Writing strips away that support. The reader may live elsewhere, read later, and know nothing of the local setting in which the prose was composed.

The lecture's rhythm depends on this beat. Without it, examples and explicit framing might sound optional, even decorative. With it, they become structural necessities. Writing is harder than speaking because the page has to carry burdens that in conversation are borne by situation, gesture, and feedback.

1.5 Generalization, Example, and the Shape of Meaning

Now the lecture narrows from large contrasts to a method we can use sentence by sentence. The interviewer proposes a principle in a deliberately sharp form: generalizations without examples and examples without generalizations are both useless. Pinker slightly softens the verdict, but he accepts the structure of the point. A generalization without a case often leaves the reader nodding without really knowing what is being claimed. An example without a generalization leaves the reader asking why the example has been introduced at all.

We may summarize the balance as

$$\text{examples} + \text{generalizations} \rightarrow \text{understanding.} \quad (1.9)$$

The interviewer then recasts the same balance as one between context and compression. That is not Pinker's exact terminology, but it is a useful editorial summary of the exchange:

$$\text{context} \leftrightarrow \text{compression.} \quad (1.10)$$

The logic unfolds in steps:

1. A generalization compresses; it sweeps over particulars.
2. Because it suppresses detail, it may fail to call up a determinate referent.
3. An example restores a concrete ballpark.
4. But an example by itself does not yet say what has been learned.
5. The generalization returns to tell us why the example matters.

Here the lecture performs something subtler than an ordinary writing lesson. It turns from exposition to semantics. Pinker gives a sequence of familiar expressions whose meanings are not recoverable simply by summing the meanings of their parts. A bathroom need not contain a bath. Going to the bathroom does not literally mean going to a room that contains one. Breakfast need not be experienced as a breaking of a fast. Christmas is ordinarily not processed as a fresh piece of compositional theology every time it is uttered.

Then the lecture widens again. Olive oil is oil made from olives, whereas baby oil is oil for babies. The same surface form supports different semantic relations. Richard Lederer's more playful examples push the lesson further. Singer transparently reflects a living rule by which "-er" turns a verb into the noun for a typical agent. Finger does not. The language preserves old formations long after their original logic has disappeared from living awareness.

Remark 1.4. Meaning is often conventional and historical rather than mechanically compositional. Pinker's point is not that no rule exists, but that the reader cannot be expected to recover intended meaning from surface form alone.

The lecture does briefly mention adultery and adulterate as an illustration of semantic drift. The prudent way to preserve that moment in notes is not to make the example carry more than it does in the lecture. The important point is that familiar words can travel far from their original relations, and that writers should never assume that formal decomposition alone yields understanding.

1.6 Style as Pleasure: Rhythm, Sound, Brevity, Humor

Once the lecture has shown that meaning is not automatically carried by words, it is ready to move into style. Pinker is careful about the order. He does not propose pleasure as a substitute for clarity. He proposes it as the next layer once clarity has been secured.

Before the lecture reaches Shakespeare and Strunk, it pauses over the freshness of children's explanations. Pinker delights in examples such as "smoke is fire vapor," because they show a mind still reaching for visible relations rather than inherited clichés. Children do not yet possess the heavy inventory of abstractions that later writers often use as a refuge. In that sense, they resemble earlier writers who had to reach for common imagery because they could not lean on a later technical vocabulary. The lesson is not that we should write like children. It is that freshness and observability are closely linked.

1.6.1 Question & Answer

Question. What makes prose not just understandable but pleasurable?

Answer. Pinker's answer is that prose has an audible life. It can move well or badly. It can carry rhythm or break it. It can grate or glide. The writer can test this directly by reading aloud, listening for friction, and revising the line until it can be spoken cleanly.

The local update rule is simple:

$$\text{read aloud} \rightarrow \text{hear friction} \rightarrow \text{revise cadence.} \quad (1.11)$$

Several stylistic variables now enter as operational quantities rather than vague compliments:

- **visual imagery:** can the sentence call up a scene?
- **metrical structure:** does the prose move in usable beats rather than stumbling?
- **sibilance:** are there too many successive hissing sounds?
- **alliteration:** is there a light recurrence of sound that gives a spark of pleasure without becoming forced?
- **brevity:** can the same meaning be delivered with less drag?

Pinker ties humor directly to this part of the lecture. Humor depends on freshness and on timing. A joke stretched too far stops being funny. A punch line advertised too heavily collapses under its own preparation. Hence the line from *Hamlet*, "brevity is the soul of wit," serves twice: it states a truth about wit, and it exemplifies that truth by its own economy.

Strunk's "omit needless words" is treated the same way. It is not merely a slogan quoted out of a handbook. It is an example of itself, and Pinker adds a practical test from his own writing life.

When a newspaper or magazine imposes a hard word limit, the forced compression often improves the prose rather than merely shortening it.

Schematically,

$$\text{needless words} \rightarrow \text{extra processing} \rightarrow \text{weaker force.} \quad (1.12)$$

The lecture's argument here is both cognitive and aesthetic. Every extra word is more work for the reader. But beyond that, compression often sharpens rhythm and forces the writer away from woolly idiom and toward concrete phrasing. Style, in this account, is disciplined pleasure.

1.7 Old Prose, Academic Prose, and LLM Prose

The lecture now broadens outward from craft to institutions, history, and technology. Pinker first explains why he cares so much about bad academic prose. The complaint is not merely stylistic fastidiousness. Bad prose wastes intelligence. It squanders public value. It forces reviewers, colleagues, and students to spend time decoding sentences that should have been made clear at the source. A paragraph that must be read five or six times before its claim emerges is not just ugly; it is inefficient and risky.

That institutional complaint then opens into a historical one. Why does older prose so often strike modern readers as vivid or even lush? Pinker offers several reasons in sequence. Writers of earlier periods were educated on strong literary models. Prose was more openly cultivated as the medium through which one presented oneself to the world. And, most importantly for this lecture's inner logic, earlier writers had fewer ready-made abstractions and clichés available to them, so they had to cast new ideas into forms that ordinary readers could picture.

The historical argument then bends toward cultural change. Pinker names a long process of informalization: less hierarchy, less ceremony, more conversational intimacy, more suspicion of anything that sounds elevated or self-consciously polished. This shift helps explain why many modern writers, even when capable of strong crafted prose, hesitate to sound too formal or too finished.

Only after all this does the lecture return to AI. That order matters. LLM prose is not introduced as an alien object; it is introduced as the latest pressure on a preexisting problem. Pinker grants its strengths first. It is often clear in the limited sense that its sentences are orderly, its syntax is plain, and its paragraphs tend to have recognizable openings and closings. But its weakness is equally striking: it is generic, prosaic, and often instantly recognizable as a pastiche of aggregate style.

We can state the contrast schematically as

$$\text{LLM prose} = \text{clear ordering} + \text{plain syntax} + \text{generic pastiche}, \quad (1.13)$$

$$\text{strong prose} = \text{clarity} + \text{concreteness} + \text{freshness}. \quad (1.14)$$

So the lecture closes with an important non-equivalence:

$$\text{clear prose} \neq \text{fresh prose}. \quad (1.15)$$

Pinker briefly speculates about why LLM prose is often more readable than the prose of academics, lawyers, or bureaucrats. Perhaps it has been hammered into shape by training and feedback. Perhaps, more speculatively, a composite of countless examples removes some of the worst local

distortions. But he does not let that speculation drift into a stronger claim than the lecture can support. He does not conclude that the human mind is simply a large language model. Instead he draws a narrower lesson: any serious account of intelligence now has to give more weight than older rule-centered views did to large-scale pattern extraction from massive input.

1.8 Summary

The lecture unfolds as a chain, and it is best read that way. It opens by rejecting the flattering story that bad prose is mainly strategic obscurity. It replaces that story with the curse of knowledge, then turns immediately to repair: real readers, real editors, real tests. From there it deepens the argument by asking what understanding is, and answers that readers need not just words but mental models. That answer leads to the asymmetry between speech and writing, and from there to the balance between example and generalization. Once the lecture has shown that meaning is neither automatic nor fully compositional, it can turn to style as something operational: rhythm, sound, compression, wit, and pleasure. Finally, the lecture widens into history, institutions, and AI, without losing the initial point. The writer's task remains the same from first page to last: to say what we mean in a form another mind can actually enter.